Fiscal Year:	FY 2011	**Task Last Updated:**	FY 12/06/2011
PI Name:	Trani, Daniela Ph.D.		
Project Title:	The Role of Intestinal Inflammation in Acute Effects Induced by Exposure to Protons During Solar Particle Events: A Mouse Model Approach		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSBRI-RFA-08-03 Postdoctoral Fellowships
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fornace, Albert (MENTOR/Georgetown University)		
Grant/Contract No.:	NCC 9-58-PF01904		
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Task Description:

POSTDOCTORAL FELLOWSHIP
Unpredictable sporadic and large SPEs may occur several times during the eleven-year solar cycle and pose serious health risks for manned exploratory missions beyond Low-Earth Orbit. These events acutely could cause the prodromal syndrome - a transient period of anorexia, nausea and vomiting that starts within a few hours, and may compromise crew performance. At expected doses, acute effects could be due to direct damage to intestinal mucosa as well as the effect of injury responses that involve intercellular signalling. Ionizing radiation (IR) activates a complex network of stress responses that affect cellular functions and cellular viability, and triggers altered expression of variety of cytokines and other intercellular messengers. Many of these signaling events can impinge on processes associated with inflammatory responses. Many studies show after IR a general over-expression of pro-inflammatory markers systemically and in tissues such as the intestine. A major mediator of inflammatory signaling is p38 MAP kinase (p38), which is upstream, and sometimes downstream, of key cytokines including TNF-a, IL-6, and IL-1ß, as well as other key mediators like COX-2 and p53. In the case of the prodromal syndrome, inflammatory signaling may impact on intestinal function either directly by damage to the mucosa, by local production of inflammatory cytokines, or by systemic effects of inflammatory cytokines on the gut. My principal hypothesis is that inflammatory signaling contributes to the prodromal syndrome and can affect the gut either by local signaling events and/or by systemic cytokine signaling. I am using a genetic approach in a defined mouse model system to study inflammatory signaling after space radiation exposure with emphasis on SPEs, and the effects of such signaling on cellular and molecular parameters in the intestine. By blocking p38 signaling with dominant-negative p38 mutant, we are able to assess the potential advantages of anti-inflammatory therapy to treat the prodromal syndrome.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our principal hypothesis is that inflammatory signaling contributes to the prodromal syndrome following exposure to radiation for therapeutic purposes or in deep space during long term missions beyond Low Earth Orbit (LEO). Radiation-induced inflammation may affect the gut either by local signaling events and/or by systemic cytokine signaling. We are using a genetic approach in a defined mouse model system to study inflammatory signaling after space radiation with emphasis on protons, and the effects of such signaling on cellular and molecular parameters in the intestine. By blocking p38 signaling with a dominant-negative p38 mutant, generated in the laboratory of my mentor Dr. Albert J. Fornace Jr., our study will contribute to assessing the potential advantages of anti-inflammatory therapies to treat the prodromal syndrome experienced by patients undergoing radiotherapy.

Task Progress:

A major mediator of inflammatory signaling is p38 MAP kinase (p38), which is upstream, and sometimes downstream, of key cytokines including TNF-a, IL-6, and IL-1ß, as well as other key mediators like COX-2 and p53. In the case of the prodromal syndrome, inflammatory signaling may impact on intestinal function either directly by damage to the mucosa, by local production of inflammatory cytokines, or by systemic effects of inflammatory cytokines on the gut. Our principal hypothesis is that inflammatory signaling contributes to the prodromal syndrome and can affect the gut either by local signaling events and/or by systemic cytokine signaling. We have employed a genetic approach in a defined mouse model system to study inflammatory signaling after space radiation exposure with emphasis on solar particle events (SPEs), and the effects of such signaling on cellular and molecular parameters in the intestine. By blocking p38 signaling with dominant-negative p38 mutant, we are able to assess the potential advantages of anti-inflammatory therapy to treat the prodromal syndrome.
We have investigated effects of both 1 Gev/n protons and 1 GeV/n 56Fe ions and have shown that in wild-type there was no enhanced GI toxicity after space radiation compared to equitoxic doses of gamma rays. We have employed doses < 2 Gy, but also higher doses that would generate a measurable effect in our animal models. Analyses of crypt survival in wt and p38DN/+ animals exposed to 3 Gy or lower doses of 1 GeV/n protons in the course of two different experiments carried out at the NSRL (in November 2009 and June 2010) are ongoing. The aim is to assess if GI toxicity in p38DN/+ mice is lower compared to wild-type animals.

For metabolomics analysis, serum and urine were collected from animals exposed to 0.5 or 3 Gy at 6, 24 hr and 1, 2, 4 weeks after irradiation; in vivo changes were measured by UPLC/TOFMS in order to identify metabolic signatures associated with the acute response to high-energy protons exposure in wt compared to p38DN/+ animals. We expect that such metabolites will reflect injury responses to this radiation and will help shed light on the underlying mechanisms involved in systemic responses and other sequela.

Bibliography Type:	Description: (Last Updated: 11/13/2018)
Abstracts for Journals and Proceedings	Datta K, Trani D, Moon B-H, Sun J, Fornace AJ Jr. "Space radiation and intestinal tumorigenesis risk assessment." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 16-19, 2010. 21st Annual NASA Space Radiation Investigators' Workshop, Abstract Book, May 2010. p. 10. , May-2010
Abstracts for Journals and Proceedings	Trani D, Datta K, Moon B-H, Sidahmed E, Sun J, Kallakury B, Fornace AJ Jr. "Enhanced intestinal tumor multiplicity and grade in vivo after HZE and proton exposure: mouse models for space radiation risk estimates." 56th Annual Meeting of the Radiation Research Society, Maui, HI, September 25-29, 2010. Abstract book. 56th Annual Meeting, Radiation Research Society, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=9e011f35-da29-44e9-81ca-4b52cf617fb4&cKey=c905d5d2-7916-4fa1-b6a7-c7c517ad48f8&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Abstracts for Journals and Proceedings	Trani D, Datta K, Moon BH, Williams E, Kallakury B, Fornace Jr AJ. "Radiation quality and enhancement of intestinal tumor multiplicity and grade in vivo." Colon Cancer in Murine Models and Humans III, Bar Harbor, Maine, October 18-22, 2010. Colon Cancer in Murine Models and Humans III, Abstract Book, October 2010. , Oct-2010
Articles in Peer-reviewed Journals	Trani D, Datta K, Doiron K, Kallakury B, Fornace AJ Jr. "Enhanced intestinal tumor multiplicity and grade in vivo after HZE exposure: mouse models for space radiation risk estimates." Radiat Environ Biophys. 2010 Aug;49(3):389-96. Epub 2010 May 20. PMID: 20490531 ; http://dx.doi.org/10.1007/s00411-010-0292-2 , Aug-2010
Awards	Trani D. "Universities Space Research Association (USRA) Young Investigator Award, February 2010." Feb-2010